Interpretable Heart Graph Learning for Brugada Syndrome Detection

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ABSTRACT —Brugada Syndrome (BrS) is a hereditary cardiac channelopathy characterized by a high risk of sudden cardiac death in otherwise healthy individuals. With a high risk of sudden cardiac death, the BrS must be detected as early as possible. An automated detection can act as an early detection of BrS. We proposed using Spatial Graph Neural Network (Spatial GNNs) rather than Histogram Gradient Boost (HistBoost), which fails to capture temporal dependency, or using 1D ResNest, which only provides temporal knowledge. We are experimenting with physio Brugada-HUCA: 12-Lead ECG Recordings for the Study of Brugada Syndrome, which contains 363 patients' recordings. In order to remove noise from the recording, we use a Butterworth bandpass filter. To avoid overfitting, signal augmentation is applied. The lead graph is constructed per patient as a weighted combination of two adjacency matrices: a fixed anatomical adjacency and a correlation adjacency. To ensure robust and unbiased generalization estimates, each model was evaluated using a 5-seed protocol (seeds 42-46). Across a robust 5-seed evaluation protocol, all three models demonstrated meaningful discriminative capability on the highly imbalanced Brugada-HUCA dataset (1:3.8 class ratio). ResNet achieves the highest overall F2-Score (0.7282) and accuracy (0.8255), while SpatialGNN achieves the highest recall (0.8500). Recall itself is the most important metric within this dataset, where a missed Brugada diagnosis can result in sudden cardiac death. HistBoost confirmed the lack when capturing complex ECG morphologies, as fortified by the HistBoost recall metric (0.6333). Through Grad-CAM visualization, we visualized which lead is the most influential. Spatial GNNs consistently mark V1 and V2 as the most influential leads in the classification decision. While ResNet shows diffusion, globally distributed activation maps that lack clinical specificity. This corroborates the Spatial GNNs to capture lead importance scores that directly support clinical decision-making.

KEYWORDS — Brugada Syndrome, 12-Lead ECG, Graph Neural Networks, Interpretability, Clinical Decision Support

I. INTRODUCTION

A. BACKGROUND

Brugada Syndrome (BrS) is a genetically determined cardiac disorder characterized by a high risk of sudden cardiac death in individuals with structurally normal hearts. The Brs are characteristically involved, with ST-segment elevation in the right precordial leads (V1-V3), which represent abnormal right ventricular repolarization [1]. Critically, the majority of the affected individuals remain undetected until a potentially fatal arrhythmic event occurs, making early detection not merely beneficial but clinically essential. Manual interpretation of Brugada patterns demands specialized electrophysiological expertise and remains prone to inter-observer variability, motivating the need for a reliable automated screening framework.

Traditional machine learning methods, such as Histogram Gradient Boosting (HistBoost), offer computational efficiency and straightforward feature importance scoring. However, they only rely on handcrafted morphological features, such as J-point amplitude and T-wave polarity, which fail to capture temporal dependencies across the 12-lead ECG signal [2]. Conversely, state-of-the-art (SOTA) deep learning models, such as 1D ResNets, can achieve high accuracy by leveraging learned temporal representations across all leads simultaneously. However, this led to “black box” models that lack the transparency needed for clinical adoption, as they provide only temporal knowledge and offer no intuitive explanation of which anatomical lead relationship drives the classification decision.

Spatial Graph Neural Networks (Spatial GNNs) emerge as a solution that bridges both gaps by combining strong discriminative performance with explicit structural modelling of the 12-lead ECG. In this work, we propose a SpatialGNN framework that explicitly models the 12-lead ECG as a graph node, with edges defined by a novel hybrid adjacency matrix that combines fixed anatomical prior knowledge with dynamic Pearson correlation, resulting in clinically grounded graph structures that neither pure deep learning nor classical methods can capture [3]. This approach not only improves the recall on the Brugada class but also generates lead importance scores that directly support clinical decision-making.

B. IMPACT AND OBJECTIVES

The primary objective of this work is to deliver a computational framework that brings hope to Brugada syndrome screening by bridging the gap between high-performance deep learning and clinical interpretability. By utilizing Spatial GNNs, our team aims for beyond the simple binary classification to a system that provides actionable insights through lead importance scores. This approach offers hope for early diagnosis in asymptomatic patients, potentially preventing catastrophic events of sudden cardiac death.

II. MATHEMATICAL FORMULATION AND PREPROCESSING

A. DATASET CHARACTERISTICS

The dataset contains a total of 363 recordings of patients, with 76 of them diagnosed with Brugada, while the other 287 are normal controls. Signals are sampled at 100Hz over

12 seconds, yielding 1200 samples per lead per patient. Each subject had 4 variables.

Table 1. Dataset Characteristics

Variable	Description	Type
patient id	Unique identifier for the subject (matches the ECG file name)	Integer
basal_pattern	Indicates whether the baseline ECG exhibits a pathological Brugada-type pattern	Binary
sudden_death	Indicates whether the subject has experienced sudden cardiac death	Binary
brugada	Diagnostic label indicating the presence of Brugada syndrome	Integer (0,1,2)

There is an imbalance in the variable, such as "basal_pattern", from 363 patients; 46 patients exhibit a pathological Brugada-type pattern. while 11 patients experienced sudden cardiac death. While the Brugada class was split into three classes, we combined classes "2" and "1" into a single class called "1" [4][5].

B. SIGNAL PREPROCESSING

On preprocessing, we apply Butterworth bandpass filtering and augment the ECG signals. Butterworth bandpass filter is a type of filter that produces a maximally flat frequency response, i.e., no passband ripple or roll-off. This filter is completely defined mathematically by two parameters: the cut-off frequency and the number of poles. The cut-off itself cuts the upper limit and lower limit [6]. Which formula is given as

$$H_{(j\omega)} = \frac{1}{\sqrt{1+(\frac{\omega}{\omega_p})^{2n}}} \qquad (1)$$

Where n is the order number, ω represents the frequency, ω_p represents the cutoff frequency, and H represents the transfer function of the bandpass [7]. Within this paper, we set the lower limit as 0,5Hz and the Upper limit set to 40Hz.

Only using 363 data, it might lead the model to overfitting, so we apply augmentation on the ECG signal, such as additive noise, time shift, and amplitude scaling. Which consecutive formula is given as

$$x'(t) = x(t) + \epsilon(t) \qquad (2)$$

$$x'(t) = x(t - r) \qquad (3)$$

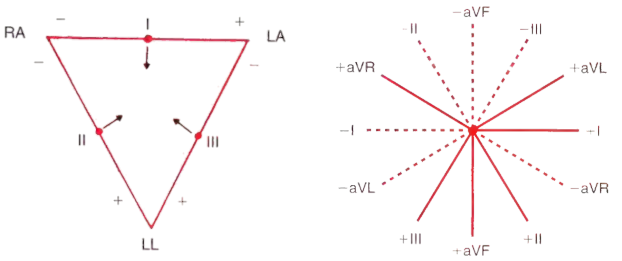
$$x'(t) = \alpha . x(t) \qquad (4)$$

where ε represents Gaussian noise, r represents a random shift, α represents amplification, and x(t) represents the actual ECG signal within the time index, while x'(t) represents the augmented ECG signal. The additive noise works by adding random Gaussian noise. This simulates real-world noise. The time shift works by shifting position, i.e., the heartbeat shifted from 0:01 to 0:02. This simulates a different starting point on recording. The amplitude scaling works by changing the signal height. It simulates a different body for each patient and electrode placement [9].

C. GRAPH CONSTRUCTION

The lead graph is constructed per patient as a weighted combination of two adjacency matrices: a fixed anatomical adjacency and a correlation adjacency. This hybrid formulation incorporates domain knowledge about electrode placement alongside learned signal correlations to produce clinically grounded graph structures

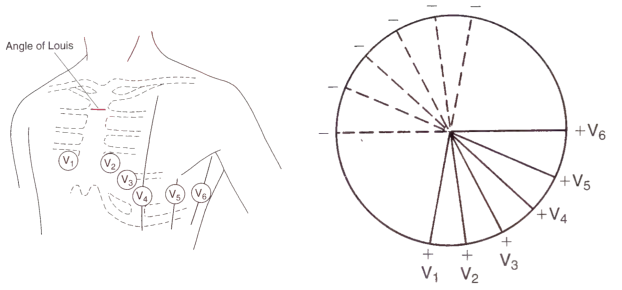
The anatomical adjacency encodes the known physical and electrical relationships between the 12 standard leads. The lead ordering used throughout is I, II, III, aVR, aVL, aVF (indices 0-5, limb leads) and V1, V2, V3, V4, V5, V6 (indices 6-11, precordial leads) [1]. Connections are defined by two clinical principles.



(a) (b)

Figure 1. (a) Einthoven's triangles. (b) a hexaxial diagram that shows the relationship of all six limb leads.

For limb leads, edges follow Einthoven's triangles (figure 1(a)), which describe the three standard bipolar leads (I, II, III) as forming a triangle of potential differences around the heart. Lead I measures the voltage difference between the right arm (RA, negative pole) and left arm (LA, positive pole), Lead II between RA and left leg (LL), and Lead III between LA and LL. The augmented unipolar leads, aVR, aVL, and aVF, are derived by referencing each limb electrode against the average of the other two. Figure 1(b) shows that the geometric regularity that makes the limb lead graph naturally representable as a structured, symmetric topology



(a) (b)

Figure 2. (a) Locations of the electrodes for the chest. (b) Spatial relationships of the six chest leads, which record electrical voltages transmitted onto the horizontal plane.

For the precordial leads, edges connect sequentially adjacent electrodes reflecting their physical placement

across the chest wall. As shown in Figure 2(a), V1 and V2 are positioned at the 4th intercostal space on either side of the sternum (located using the Angle of Louis as the anatomical landmark for the 2nd rib), V3 and V4 transition across the anterior chest, and V5 and V6 continue laterally along the axillary lines. Figure 2(b) illustrates how each precordial lead axis radiates from a common central reference point, sweeping from roughly inferior-right (V1) to lateral-left (V6) in the horizontal plane. Hence, justifying sequential edge structure [1].

The Correlation adjacency is computed using the Pearson correlation matrix of 12 lead signals, capturing the dynamic inter-lead relationship of each cardiac anatomy and pathology. However, sometimes correlation can occur by chance rather than an actual connection [8]. These two then combine as follows

$$hybrid_adj = \alpha * \beta + (1 - \alpha) * \gamma \quad (5)$$

where α represents anatomic weight, β represents anatomical adjacency, and γ represents correlation adjacency.

III. PROPOSED METHODOLOGY

A. ARCHITECTURE DESIGN

The proposed methodology for the SpatialGNN uses a combination of temporal feature extraction with spatial relationship modeling. Each lead from a 12-lead ECG is being processed through a shared per-lead ResNet decoder to create a latent temporal embedding. These features are then processed through a graph convolution (GCN or GAT) to aggregate the spatial information across the lead graph.

We construct the models as a multi-task model to prevent overfitting and enrich the latent temporal embeddings. Although the primary objective is Brugada syndrome classification, joining these clinically correlated targets could strictly regulate the shared representation and force the model to learn robust morphological features rather than spurious noise. It should be noted that this only works in Deep Learning since multitasking in traditional models works independently of each other.

A hyperparameter search using 3-seed random search (seeds 42, 43, 44), similar to the Monte Carlo protocol, to identify stable configurations across different initialization weights and data splits. Final comparison between different architectures was then performed using a 5-seed evaluation (seeds 42-46), where two entirely unseen seeds provided an unbiased estimate of generalization. The best configuration from this is summarized below:

1. **HistBoost:** Max depth 3, minimum 40 samples per leaf, class weights balanced, and 200 iterations with balanced class weights. Auxiliary tasks' basal patterns were enabled.
2. **ResNet:** Learning rate of 0.0015, weight decay of 0.0005, and 0.5 dropout. The architecture utilized channel dimensions of [32, 64, 128, 256]. Auxiliary tasks' sudden death risk was enabled.
3. **SpatialGNN**: Learning rate of 0.0012 with 0.01 weight decay and 0.5 dropout. The resnet channel dimension of [32, 64, 128, 256]. The graph component utilized 3 GAT layers with a hidden dimension of 64, a 0.6 correlation threshold, and a 0.7 anatomical weight. Auxiliary tasks' basal patterns were enabled.

B. LOSS FUNCTION & OPTIMIZATION

The dataset contains a significant class imbalance (1:3.8 ratio). To address this, we implemented two strategies. First, a weighted random sampler was used during training to oversample the positive Brugada cases within each batch. Second, we utilized focal loss (FL) [10] for all tasks to concentrate gradient updates on hard examples. The overall training objective is formulated as a weighted sum of the primary and auxiliary losses:

$$L_{total} = FL_{target} + \lambda(\Sigma FL_{auxiliary\ feature}) \quad (6)$$

$$FL(p_t) = -\alpha_t(1 - p_t)^{\gamma} log(p_t) \quad (7)$$

Where the $\alpha = 0.79$ derived from the inverse class frequency (287/363) and $\gamma = 2.0$ downweight easy negatives and concentrate gradient updates on hard and minority-class examples. At the same time, the auxiliary weight λ was set to 0.3 to prioritize the primary task (Brugada).

All deep learning models are trained with AdamW, a cosine annealing learning rate schedule with optional linear warmup, and early stopping on the validation F2-score. This metric was selected specifically for clinical safety, as it weights recall twice as heavily as precision, reflecting the reality that a missed Brugada diagnosis carries far greater clinical cost than an unnecessary cardiology referral.

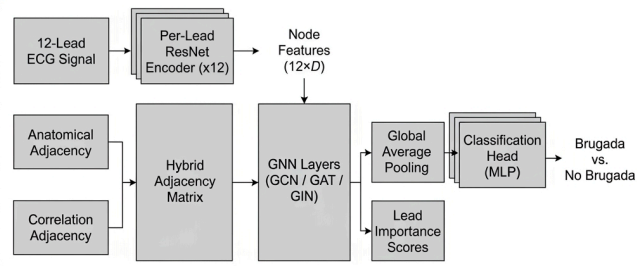


Figure 3. SpatialGNN Pipeline

IV. EXPERIMENTAL RESULTS AND EVALUATION

A. PERFORMANCE METRICS

Three models were evaluated using a 5-seed protocol (seeds 42-46) to ensure robust and unbiased generalization estimates. Table 2 presents the complete quantitative results on the held-out test set

Table 2. Performance of Each Model in the 5-seeds Validation

Model	Accuracy	Precision	Recall
HistBoost	0.7855 ± 0.0506	0.5234 ± 0.1104	0.6333 ± 0.0408
ResNet	**0.8255 ± 0.0374**	**0.5853 ± 0.0841**	0.7833 ± 0.1000

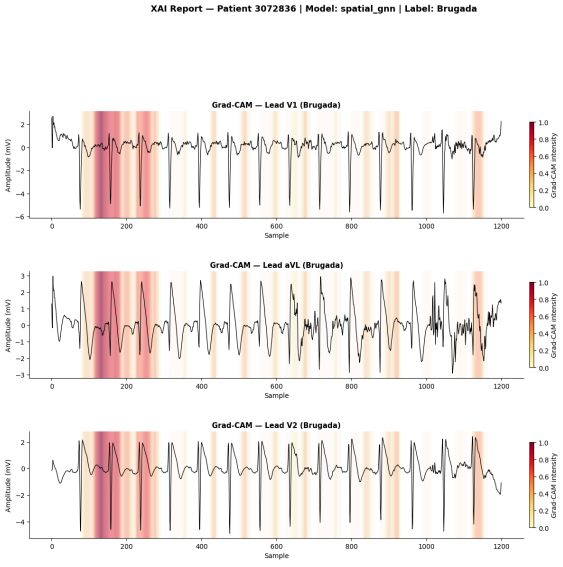
Spatial GNN	0.7455 ± 0.0905	0.4791 ± 0.0958	**0.8500 ± 0.1616**

Model	F1-Score	ROC-AUC	F2-Score
HistBoost	0.5686 ± 0.0724	0.7872 ± 0.0340	0.6044 ± 0.0496
ResNet	**0.6630 ± 0.0575**	**0.8713 ± 0.0439**	**0.7282 ± 0.0721**
Spatial GNN	0.5996 ± 0.0866	0.8465 ± 0.0728	0.7212 ± 0.1039

The primary evaluation metric is F2-Score, which weights recall twice as heavily as precision, reflecting the clinical priority of minimizing missed Brugada diagnoses; a missed diagnosis carries greater clinical risk than a false referral. Under this metric, ResNet achieves the highest F2-Score of 0.7282, followed closely by SpatialGNN at 0.7212. Notably, SpatialGNN achieves the highest recall of 0.8500 across all models, indicating that it misses the fewest true Brugada cases, the most clinically critical property for a screening tool. HistBoost performs lowest across all metrics, particularly in recall (0.6333), confirming that handcrafted morphological features are insufficient to capture the full complexity of Brugada ECG patterns [11].

B. CLINICAL INTERPRETABILITY

Beyond classification performance, the SpatialGNN uniquely provides clinical interpretability through two complementary XAI mechanisms, visualised in **Figure 4** (SpatialGNN) and **Figure 5** (ResNet) for Patient 3072836 (Brugada-positive).



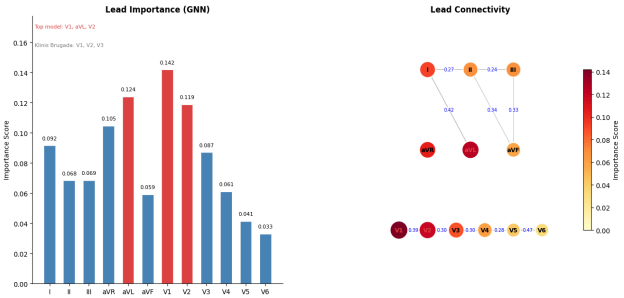


Figure 4. XAI Report for SpatialGNN Patient 3072836 (Brugada). Top: Grad-CAM activation maps for top-3 leads. Bottom-left: GNN lead importance scores. Bottom-right: Lead connectivity graph.

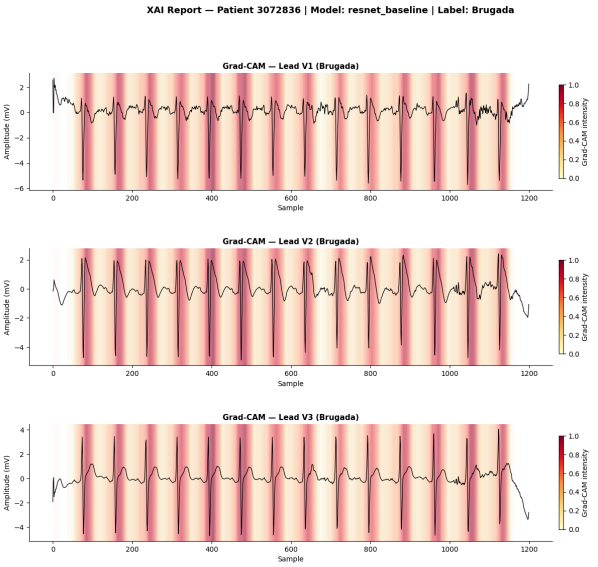


Figure 5. XAI Report for ResNet — Patient 3072836 (Brugada). Grad-CAM activation maps for leads V1, V2, and V3.

Post-hoc XAI methods applied to deep ECG models can be evaluated both at the local (per-sample attribution) and global (aggregated across patient subgroups) levels, enabling both individual-level diagnosis support and population-level knowledge discovery. In this work, we apply Grad-CAM at the local level for per-patient visualization.

For the ResNet baseline (Figure 5), Grad-CAM activation maps across leads V1, V2, and V3 exhibit a diffuse, globally distributed pattern spanning nearly the entire 12-second recording. This behavior is consistent with findings that signal-form explainability maps can fail to emphasize clinically relevant regions in isolation, whereas complementary spatial representations may successfully identify pathological patterns. While this enables strong classification performance (AUC 0.8713), the uniform activation offers limited clinical utility, as clinicians cannot identify which specific morphological features drove the decision.

In contrast, the SpatialGNN (Figure 4) produces markedly more focused Grad-CAM activations. The high-intensity regions are concentrated in the early portion of each cardiac cycle, consistent with the QRS-ST junction, where the coved-type ST-segment elevation characteristic of Brugada syndrome typically manifests. Deep neural networks applied to ECG data are increasingly capable of accurately identifying cardiac conditions and hidden clinical factors,

though their black-box nature remains a common concern that XAI methods are employed to address

The GNN lead importance scores (Figure 4, bottom-left) further reinforce this interpretability advantage. V1 achieves the highest importance score (0.142), followed by aVL (0.124) and V2 (0.119). The prominence of V1 and V2 directly aligns with established clinical criteria, where ST-segment elevation in the right precordial leads V1–V3 constitutes the primary diagnostic hallmark of Brugada syndrome [12]. The Lead Connectivity graph (Figure 4, bottom-right) further demonstrates that precordial leads form a tightly interconnected subgraph, suggesting the GNN has implicitly learned the anatomical proximity of the right precordial lead group. Together, these findings demonstrate that the SpatialGNN produces explanations that are anatomically coherent and clinically actionable, a critical requirement for deployment in cardiac screening pipelines. Full XAI reports, including HGB feature importance and False Negative comparisons, are available at the public repository [13].

V. CONCLUSION

This study presented an interpretable computational framework for automated Brugada syndrome detection from 12-lead ECG signals, comparing three model architectures: Histogram Gradient Boosting (HistBoost), 1D ResNet, and Spatial Graph Neural Network (SpatialGNN). Across a robust 5-seed evaluation protocol, all three models demonstrated meaningful discriminative capability on the highly imbalanced Brugada-HUCA dataset (1:3.8 class ratio).

The quantitative results reveal a clear trade-off between accuracy and clinical safety. ResNet achieves the highest overall F2-Score (0.7282) and accuracy (0.8255), while SpatialGNN achieves the highest recall (0.8500), the metric most critical for clinical screening, where a missed Brugada diagnosis can result in sudden cardiac death in an otherwise asymptomatic patient. HistBoost, despite its computational efficiency, lags significantly in recall (0.6333), confirming the limitations of handcrafted feature extraction for capturing complex ECG morphologies.

Beyond raw performance, the primary contribution of this work lies in the interpretability of the SpatialGNN. Through Grad-CAM visualization, the model demonstrably focuses its attention on the QRS-ST junction region of each cardiac cycle, precisely the anatomical site of coved-type ST-segment elevation that defines Brugada syndrome according to international consensus criteria [1]. In contrast, the ResNet baseline produces diffuse, globally distributed activation maps that lack clinical specificity. Furthermore, the GNN lead importance scores consistently rank V1 and V2 as the most influential leads in the classification decision, directly corroborating established electrophysiological knowledge that right precordial leads carry the primary diagnostic signal in Brugada syndrome [1]. This convergence between model-derived importance and clinical knowledge provides meaningful evidence that the SpatialGNN has learned physiologically grounded representations rather than spurious correlations.

This work has several limitations. The dataset is limited to 363 patients from a single center in Spain, which may constrain generalizability to other populations and ECG acquisition systems. Additionally, the 12-second recording duration may not capture transient Brugada patterns that manifest intermittently. Future work should validate the proposed framework on larger, multi-center datasets, explore dynamic graph construction strategies that adapt to patient-specific lead correlations, and conduct prospective clinical evaluation to assess the real-world utility of the generated XAI outputs as decision-support tools for cardiologists.

REFERENCES

[1] Goldberger, A. L., Goldberger, Z. D., & Shvilkin, A. (2013). Goldberger's Clinical Electrocardiography: A Simplified Approach (8th ed.). Elsevier Saunders. ISBN: 978-0-323-08786-5.

[2] Sethi, A., et al. (2025). "ProtoECGNet: Case-Based Interpretable Deep Learning for Multi-Label ECG Classification with Contrastive Learning." arXiv preprint arXiv:2504.08713.

[3] Ma, Y., et al. (2026). "BEAT-Net: Injecting Biomimetic Spatio-Temporal Priors for Interpretable ECG Classification." arXiv preprint arXiv:2601.07316.

[4] N. Costa Cortez and D. Garcia Iglesias, "Brugada-HUCA: 12-Lead ECG Recordings for the Study of Brugada Syndrome," version 1.0.0, PhysioNet, 2026, RRID: SCR_007345. [Online]. Available: https://doi.org/10.13026/0m2w-dy83

[5] Goldberger, A., Amaral, L., Glass, L., Hausdorff, J., Ivanov, P. C., Mark, R., ... & Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. Circulation [Online]. 101 (23), pp. e215–e220. RRID: SCR_007345.

[6] P. Podder, M. M., Hasan, M. R. Islam, and M. Sayeed, *"Design and Implementation of Butterworth, Chebyshev-I and Elliptic Filter for Speech Signal Analysis,"* arXiv preprint arXiv:2002.03130, 2020. [Online]. Available: https://doi.org/10.48550/arXiv.2002.03130

[7] Electronics Tutorials, "Active Band Pass Filter: Frequency Response of an Active Band Pass Filter," Electronics Tutorials, 2026. [Online]. Available: https://www.electronics-tutorials.ws/filter/filter_8.html

[8] X. Luo, J. Wu, J. Yang, S. Xue, A. Beheshti, Q. Z. Sheng, D. McAlpine, P. F. Sowman, A. Giral, and P. S. Yu, "Graph Neural Networks for Brain Graph Learning: A Survey," *arXiv preprint arXiv:2406.02594*, 2024.

[9] Raghu, A., Shanmugam, D., Pomerantsev, E., Guttag, J., & Stultz, C. M. (2022). "Data Augmentation for Electrocardiograms." Proceedings of the Conference on Health, Inference, and Learning. pp. 282-310.

[10] Lin, T. Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). "Focal loss for dense object detection." Proceedings of the IEEE international conference on computer vision. pp. 2980-2988.

[11] P. Wagner, T. Mehari, W. Haverkamp, and N. Strodthoff, "Explaining deep learning for ECG analysis: Building blocks for auditing and knowledge discovery," Computers in Biology and Medicine, vol. 176, p. 108525, 2024. https://doi.org/10.1016/j.compbiomed.2024.108525

[12] F. Murat Duranay, E. Murat, Ö. Yıldırım, et al., "MM-GradCAM: an improved multimodal GradCAM method with 1D and 2D ECG data for detection of cardiac arrhythmia," Scientific Reports, vol. 16, no. 7919, 2026. https://doi.org/10.1038/s41598-026-38654-w

[13] K. Shridhara, E. A. S. Ramadhan, and I. A. Setyawan, "Interpretable Heart Graph Learning for Brugada Syndrome Detection, Source Code and Extended XAI Reports," GitHub, 2026. [Online]. Available: https://github.com/kiuyha/idsc2026-brugada